Syllabus

1 Using the Handouts

Each handout comes with about 30 problems, chosen to illustrate different ideas.

- Each one has a point value from **1** to **5**, reflecting the amount you'll learn from doing it. (Some **2** point problems are short but tricky, while some **5** point problems are long but straightforward.)
- Problems marked with **[A]** require more advanced mathematical techniques. These problems are less relevant to Olympiad physics but are chosen to demonstrate interesting things.
- For multi-part problems, particularly hard subparts are marked with stars ($\star$).
- To give you some test-taking practice, some problems are marked with a clock. I recommend doing them in one setting with only pencil, paper and a scientific calculator, and writing a solution by hand like in an actual exam. Usually the time limit matches the competition (e.g. 10 minutes per point in the IPhO or APhO) but sometimes it'll be shorter if the question is short.

2 Practice Advice

For USAPhO preparation, try about 2/3 of the total points, and all of the USAPhO problems. It's also valuable to try problems from all competitions. The average IPhO question is longer than a USAPhO problem, but the time limits are often more generous. Conversely, newer USAPhO problems often introduce new ideas quickly, and can be valuable practice even for IPhO contestants outside the US. Serious IPhO contestants, aiming for gold medals, should try almost everything.

It is extremely important not to waste problems by reading their solutions too fast. Once you know a problem's solution, you can never benefit from it the same way again. If you're stuck, with no idea how to begin, try peeking at one line of the solution. If this happens often even for the easier problems in a handout, I recommend spending some hours doing background reading from the textbooks listed at the top of each handout, and doing simpler problems from them. (The handouts will also list some fun, but generally optional supplemental reading.)

Another common scenario is that you've made some progress, but aren't sure which of several candidate answers is right. (If this never happens to you, you're probably jumping to conclusions too quickly!) In this case, I strongly encourage you to think harder, and settle on a single final answer. Also, you should thoroughly understand your own logic before consulting the handout solution, e.g. by writing a clean solution sketch. The reason is that this will train you to distinguish valid and subtly wrong arguments. This skill is far more important than just getting the right answer, and it will continue paying off long after you're done with Olympiad physics.

3 Accessing Problems

Most exams, such as the IPhO, APhO, EuPhO, IZhO, USAPhO, NBPhO (formerly EFPhO), INPhO, AuPhO, and BAUPC are collected on **phoXiv**. For errors in official solutions, see Stefan Ivanov's errata list, which will be referred to as needed. If an old link doesn't work, try the Internet Archive.

4 Textbooks and Resources

As a foundational prerequisite, I recommend Halliday, Resnick, and Krane, *Physics* (5th edition). In my experience coaching the American team, you can get an IPhO gold medal in the 2020s using just the content in this book. It is also possible to start the handouts after any other good calculus-based introductory textbook, though you might need a bit more problem solving practice at the start.

Though more advanced books aren't strictly necessary, it is both valuable and fun to learn physics more deeply. Throughout the handouts, I will reference chapters from textbooks for optional supplemental reading. The most important books are:

- Kleppner and Kolenkow, *An Introduction to Mechanics*. Used at MIT, written more like a physics book. Has good problems with a practical emphasis. The 1st and 2nd editions have almost the same problems and content, so you can use either. The 2nd edition splits up some of the early chapters; problem number references in the handouts will be to the 1st edition.
- Morin, *Mechanics*. Used at Harvard, written more like a math book. Has a large stock of elegant and tricky, albeit sometimes contrived mechanics problems, with less relevance to real-world physics. Also contains an excellent, careful introduction to special relativity.
- Purcell and Morin, *Electricity and Magnetism*. Does electromagnetism with vector calculus and relativity baked in. Famous for using relativity to motivate magnetism, rather than just postulating it. Has well-written problems that provide insight. The 3rd edition is a substantial improvement on the 2nd, with SI units adopted throughout and more challenging problems. (Griffiths is an excellent undergraduate electromagnetism book, but I don't recommend using it for Olympiads, since many of its problems use vector calculus techniques beyond the Olympiad syllabus. However, the handouts will contain some good, elementary Griffiths problems.)
- *The Feynman Lectures on Physics*. A wonderful source of physical insight. Most problem sets will have some chapters assigned for entertainment and enrichment. (But it is not a replacement for a standard introductory text; one should learn the material in Halliday and Resnick first.)
- Wang and Ricardo, *Competitive Physics*, used by the Singapore physics team. This is a very good supplementary text, containing clear, detailed explanations of precisely the theory which appears on the IPhO but not in a standard introductory textbook. It is especially useful for thermodynamics and waves. On the other hand, its problems tend to be a bit mathematically complicated or contrived, and there are a decent number of typos in the solutions.

Even more advanced textbooks, at the mid-undergraduate level, are listed in my second advice file, but they are generally not helpful for Olympiads. Finally, besides past Olympiads and textbooks, problems are also sourced from the following books, which you can consult for additional practice.

- ⋆ *200 Puzzling Physics Problems* and *200 More Puzzling Physics Problems*. Brief and tricky questions, often drawn from the excellent, long-running Hungarian Eötvös competition. The first book is highly recommended; the second book is at times too mathematically contrived to be too relevant to Olympiads, but still lots of fun.
- ⋆ Handouts by Jaan Kalda. These handouts and formula sheets provide excellent training for Eastern European style Olympiads. Good solutions written by students are available here. (If you like the style of the NBPhO, you can find more questions from earlier rounds here. We

won't use them in the handouts, but they're more straightforward than NBPhO problems and can serve as a good stepping stone.)

- Cahn and Nadgorny, *A Guide to Physics Problems.* A thorough collection of graduate school qualification exam problems. Many great classic problems were given on these exams, though most problems in the book are too technical to be useful for Olympiad preparation. The massive series entitled "Major American Universities Ph.D. Qualifying Questions and Solutions", edited by Yung-Kuo Lim, has more questions of this type, though they're easier on average.
- Here are some cute, ~ 100 page books which you might enjoy spending an afternoon with.
  - Mahajan, *Order of Magnitude Physics*. A nice book about dimensional analysis and estimation. *The Art of Insight* is a longer work by the same author on the same themes.
  - Lemons, *A Student's Guide to Dimensional Analysis.* Shows a good variety of examples throughout physics, with interesting historical asides.
  - Levi, *The Mathematical Mechanic.* Uses mechanical setups to find slick solutions for calculus and geometry problems.
- There are a number of compilations of Russian physics problems.
  - Irodov, *Problems in General Physics.* This is a massive list of 2,000 practice problems. Personally, I don't recommend using it for the USAPhO or IPhO, because there are too many tedious filler problems, and only the briefest of solutions. It was a great book 40 years ago because there was nothing with comparable scope, but we have better alternatives now.
  - Krotov, *Problems in Physics.* A collection of good, old Russian Olympiad problems.
  - Савченко, Задачи по Физике ("Savchenko"). An updated and upgraded version of Irodov, with many interesting problems and less filler. It is used for training in Eastern Europe, but it has not been translated from Russian. It shares many problems with Kalda's handouts.
  - Kiselev and Slobodyanin, *Russian Physics Olympiads.* The modern analogue of Krotov, with problems from 2005 to 2017. Most of the tricky homework problems in PhysicsWOOT are drawn from it. Similar style to NBPhO problems, but slightly more difficult.

  There's also a great tradition of tough Russian problems at the university level, though unfortunately many have not been translated from Russian. For some problems accessible to first year students, see here. The Landau and Lifshitz series is also great for further study.
- In the United Kingdom, there is a tradition of challenging physics exams at university (dating back to the Cambridge Tripos). Below are some books of problems at the early university level.
  - Povey, *Professor Povey's Perplexing Problems.* A collection of simple but tricky Oxford admissions interview questions with neat historical anecdotes.
  - Pippard, *Cavendish Problems in Classical Physics.* Some classic problems used for second year exams in Cambridge.
  - Thomas and Raine, *Physics to a Degree.* A collection of well-motivated questions used for undergraduate physics training, with many real-world applications.
- There is a great deal of interesting Olympiad material from East Asian and South Asian countries, though many such resources have not been translated to English.

- *Physics Olympiad – Basic to Advanced Exercises*, training material used by the Japanese physics team. This book contains clear explanations of basic theory, along with a useful introduction to experiments.
- H.C. Verma, *Concepts of Physics*. This book has decent theory, and many worked examples and problems. If you're preparing for the Physics Olympiad in India, it's a good place to start, because it's comprehensive and adapted to the Indian physics curriculum. (For example, compared to HRK it has more material on optics, and includes more applied topics such as electrolysis, photometry, humidity, calorimetry, and cathode ray tubes.) Its coverage of theory is occasionally unclear, especially in the later chapters, because it sometimes throws out magic formulas and recipes without explaining what they mean or why they work. But despite these flaws, it's still a lot better than most Indian introductory books.
- For Olympiads, I advise against using JEE cram books by Cengage or Arihant. The theory in these books is extremely brief; they essentially list a bunch of random formulas, some of which are misleading or meaningless. The problems are short and largely rely on applying single tricks or memorized results, and the solutions often don't make sense. These books aren't written to encourage the deeper thought required to solve Olympiad problems. Some top Indian students don't even recommend them for JEE! If you do use a JEE book, the best is probably *Pathfinder*, which has a decent selection of problems.
- 舒幼生, 物理学难题集萃. A classic compilation of problems from the former head coach of the Chinese physics team, with full solutions, comparable in scope to Savchenko. Contains a wide variety of tough and unique problems, though some are unnecessarily messy.
- 郑永令, 国际物理奥赛的培训与选拔. A slim book by another former head coach, with a wide range of great practice problems with full solutions. There is absolutely no filler: all of the questions are interesting, with the hardest reaching beyond IPhO level. This is one of the top books used today by serious contenders for the Chinese IPhO team.

Finally, a quick note on referencing sources. There isn't an infinite supply of clean, insightful Olympiad-level physics problems. If you exclude trivial variations or pointless complications, I would estimate only a few thousand problems are possible, and many of those are classics, which are decades or even centuries old, and have appeared in dozens of books, papers, and competitions. I try to cite sources as much as possible, not because those are actually the original sources, but because they might contain helpful alternative solutions. However, I also often modify questions, by correcting errors, adding new subparts to illustrate subtleties, or removing explicit statements of things that you can figure out for yourself. So the sources I cite should serve as only a rough guide; sometimes my solutions differ substantially from theirs.

5 Curriculum Outline

For each handout, the relevant material in Halliday, Resnick, and Krane is a prerequisite, and most handouts require all of the previous ones in that topic. Prior exposure to vector calculus can be helpful for thermodynamics and electromagnetism, but not necessary.

- 2 units of problem solving.
  - **P1**: dimensional analysis, limiting cases, series expansions, differentials, iterative solutions.
  - **P2**: probability, error analysis, data analysis, estimation.

- 8 units of mechanics.
  - **M1**: kinematics. Solving $F = ma$, projectile motion, optimal launching. (**P1** helpful)
  - **M2**: statics. Force and torque balance, extended bodies, pressure and surface tension.
  - **M3**: dynamics. Momentum, energy and center-of-mass energy, collisions.
  - **M4**: oscillations. Damped/driven oscillators, normal modes, small oscillations, adiabaticity.
  - **M5**: rotation. Angular kinematics, angular impulse, physical pendulums. (**P2** helpful)
  - **M6**: gravity. Kepler's laws, rocket science, non-inertial frames, tides. (**E1** helpful)
  - **M7**: fluids. Buoyancy, Bernoulli's principle, viscosity and surface tension. (**M2** helpful)
  - **M8**: synthesis. 3D rotation, precession, and tricky problems.
- 3 units of thermodynamics.
  - **T1**: ideal gases, statistical mechanics, kinetic theory, the atmosphere. (**M7** required)
  - **T2**: laws of thermodynamics, quantum statistical mechanics, radiation, conduction.
  - **T3**: surface tension, real fluids, phase transitions, compressible flow.
- 8 units of electromagnetism.
  - **E1**: electrostatics. Coulomb's law, Gauss's law, potentials, conductors.
  - **E2**: electricity. Images, capacitors, conduction, DC circuits.
  - **E3**: magnetostatics. More circuits, Biot–Savart law, Ampere's law, dipoles and solenoids.
  - **E4**: Lorentz force. Dynamic charges, permanent magnets, solid state physics. (**M4** helpful)
  - **E5**: induction. Faraday's law, inductors, dynamos, superconductors.
  - **E6**: circuits. RLC circuits, filters, normal modes, diodes. (**M4** required)
  - **E7**: electrodynamics. More circuits, displacement current, radiation, field energy-momentum.
  - **E8**: synthesis. Electromagnetic fields in matter, and tricky problems.
- 3 units of relativity.
  - **R1**: kinematics. Lorentz transformations, Doppler effect, acceleration, classic paradoxes.
  - **R2**: dynamics. Momentum, energy, four-vectors, forces, relativistic strings. (**E4** helpful)
  - **R3**: fields. Electromagnetic field transformations, the equivalence principle. (**E7** required)
- 3 units of waves.
  - **W1**: wave equation, standing waves, music, interferometry. (**M4** required)
  - **W2**: interference and diffraction, crystallography, real world examples. (**E7** required)
  - **W3**: sound waves, water waves, polarization, geometrical optics. (**M7** required)
- 3 units of modern physics.
  - **X1**: semiclassical quantum mechanics, bosons and fermions. (**M4**, **T2**, **W1** required)
  - **X2**: nuclear, particle, and atomic physics. (**R2** required)
  - **X3**: condensed matter, astrophysics, and cosmology. (**W3** helpful)

- Miscellaneous units and practice exams.
  - Review for mechanics (**MRev**), electromagnetism (**ERev**), and everything else (**XRev**).
  - 3 practice USAPhOs focused on mechanics, electromagnetism, and everything else.
  - 4 general practice USAPhOs.
  - Extra theoretical topics, and a list of unused theoretical exams for practice.
  - Advice for preparing for the IPhO experimental exam.

My recommended path through the 24 core handouts (underlined above) is **P1**, **P2**, **M1–4**, **E1–4**, **M5–7**, **T1–3**, **E5–7**, **R1**, **R2**, **W1**, **W2**, **X1**. This reflects the sophistication of the problems, and also splits up the long topics. The other units are more relevant for IPhO preparation.

For USAPhO preparation, I recommend trying all the core handouts, but with less emphasis on the hardest 1/3 of the problems. If you start in September, a good pace is one handout per week; if you start in summer, a good pace is one handout per 1.5 weeks. Note that later handouts tend to require more background reading and have more complex questions.

Topics fluctuate significantly from year to year, but I would estimate that on average, a bit more than half of the points on the USAPhO and IPhO are devoted to mechanics and electromagnetism, with the rest roughly evenly divided between relativity, thermodynamics, waves, and modern physics. Of these four special topics, thermodynamics and relativity tend to be a bit more common at the USAPhO. At the IPhO, problems about specific advances in modern physics are common, though many of the parts in such problems boil down to mechanics or electromagnetism.